

MatchPoint AI

Week 09 Progress & Integration Report — Course: CSE4104 (Section 7A, Team T05)

Project: MatchPoint AI (AI Resume & ATS Platform)	Submission Date: August 19, 2026
Course: CSE4104 Web App Development	Doc ID: CSE4104-7A-T05_Week09Progress

1. Team Identification & Responsibilities

Member Name	Student ID	Academic Role	Core Development Responsibilities
Junnatul Farhana	11230121088	Team Leader	Project Management, SRS Governance, QA & Workflow Integration
Farhana Azgar Orin	11230121070	Frontend Lead	React UI Architecture, Radial ATS Gauge HUD, Themes & Responsive UX
Victor Mallick	11210320676	Backend Lead	Express REST APIs, In-Memory PDF/DOCX Parsers, Client-Server Wiring
Easin Mohammad Rayhan	11220120789	Database & AI Lead	PostgreSQL / Supabase Schema & RLS, Gemini AI Failover Pool

2. Feature Completion Checklist & Status Matrix

Feature / Module	SRS Ref	Status	Implementation Summary
Authentication & RBAC	SRS §2.1	Completed (100%)	JWT session auth, password eye toggles, candidate vs. admin route guards.
Resume Parser Engine	SRS §2.2	Completed (100%)	In-memory PDF (pdfjs-dist) & DOCX (mammoth) binary text extraction.
Job Description Ingestion	SRS §2.3	Completed (100%)	Role, company, and JD text parsing with client/server validation.
ATS Score & Keyword Gap	SRS §2.4	Completed (100%)	Radial Target HUD Gauge + Matched vs. Missing competency breakdown.
Bullet Point Optimizer	SRS §2.5	Completed (100%)	Interactive side-by-side STAR-framework AI rewrites with 1-click copy.
Interview Question Bank	SRS §2.6	Completed (100%)	Technical, Behavioral & HR categories with Recruiter Intent context.
Dual-Market Job Matching	SRS §2.7	Completed (100%)	Dual-market: Bangladesh Tech & Global Remote with direct LinkedIn links.
History & Profile	SRS §2.8	Completed (100%)	Timestamped analysis records on /history and editable career profile.
Admin Operations Panel	SRS §2.9	Completed (100%)	User monitoring, metrics, AI token tracking, service health, audit logs.
AI Multi-Key Failover Pool	Week 08 Plan	Completed (100%)	Auto-failover across 3 Gemini API keys + deterministic NLP fallback.

3. End-to-End User Workflows & API Integration

End-to-End Workflow: Visitor Register → Login → Dashboard → Upload Resume & Job Description → Server-side Parsing → Gemini AI Multi-Key Evaluation → Radial ATS Result Display → STAR Bullet Optimizer → Interview Question Bank → Dual-Market Job Matching.

Method	Endpoint	Controller	Status
GET	/api/health	server.js	200 OK (Configured)
POST	/api/auth/register & login	authController.js	200 OK (JWT Issued)
POST	/api/upload	uploadController.js	200 OK (PDF/DOCX Parsed)
POST	/api/analysis/gap-analysis	analysisController.js	200 OK (ATS Score Computed)
POST	/api/interview/generate	interviewController.js	200 OK (18 Questions)
POST	/api/jobs/recommendations	jobsController.js	200 OK (6 BD + 4 Abroad)
GET	/api/user/history	userController.js	200 OK (Records Retrieved)
GET	/api/admin/analytics & logs	adminController.js	200 OK (RBAC Verified)

4. AI Architecture & Fault-Tolerant Error Handling

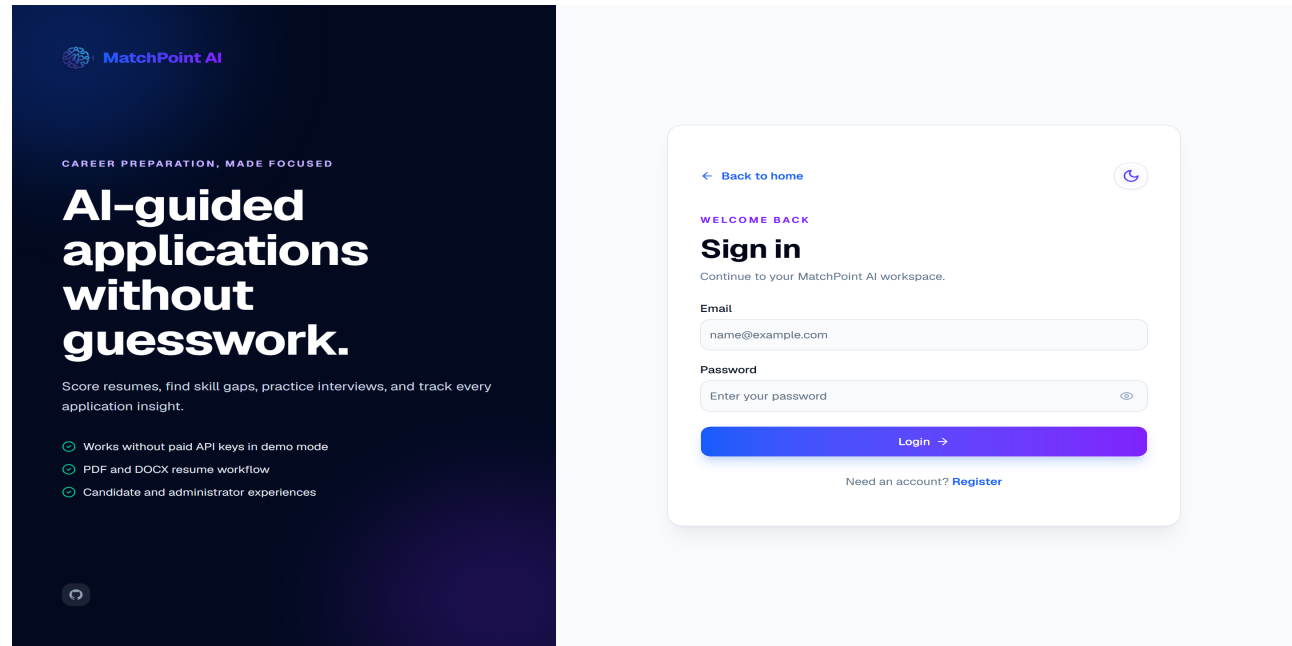
- **Google Gemini Pool:** Auto-rotates across 3 API keys upon HTTP 429 rate limit events.
- **Deterministic NLP Fallback:** Executes local resume keyword scoring if external APIs time out.
- **Edge-Case Protection:** Blocks non-resume files with a 0% ATS score and explicit warning alert.

5. Project Deliverables & Repository Links

- **GitHub Repository:** <https://github.com/victormallick/cse4104-7a-t05-matchpointai>
- **Postman API Collection:** [postman/MatchPoint_AI_APIs.postman_collection.json](#)
- **Database Schema SQL:** [database/schema.sql](#) (8 PostgreSQL production tables)

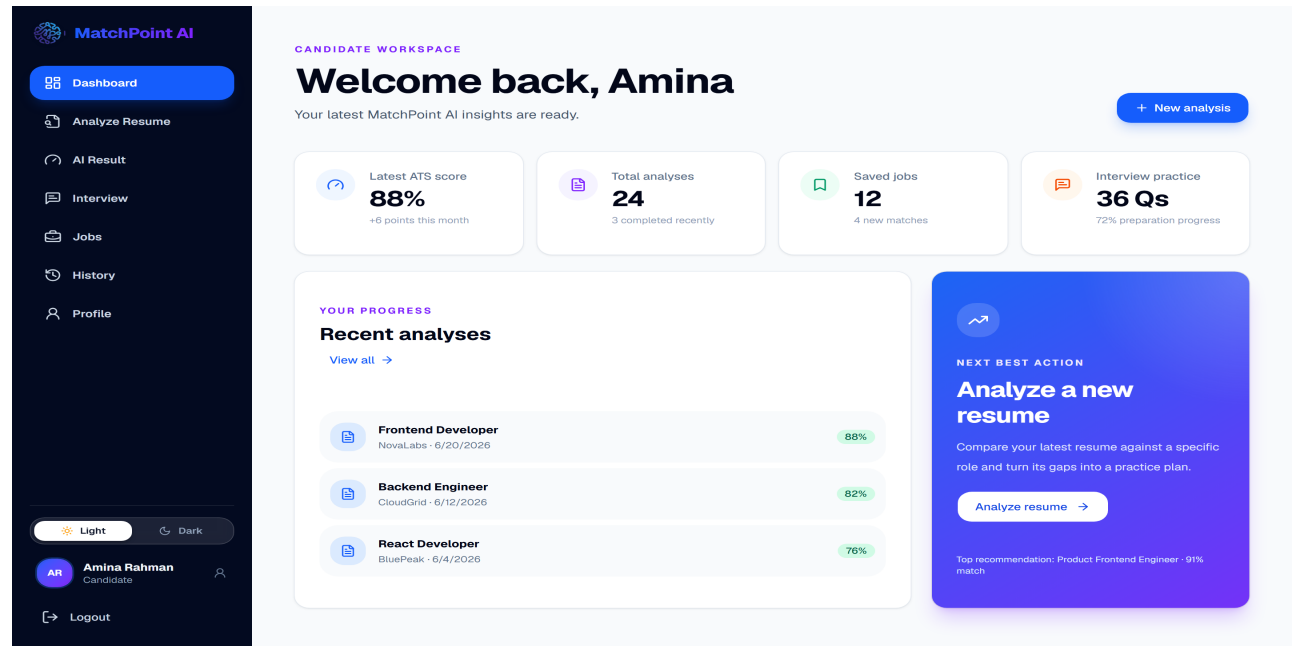
6.1 Visual Verification — Live System Screenshots (1×1 Format)

Figure 1: Authentication & Password Toggle



Sign-in and registration interface with interactive password visibility eye buttons.

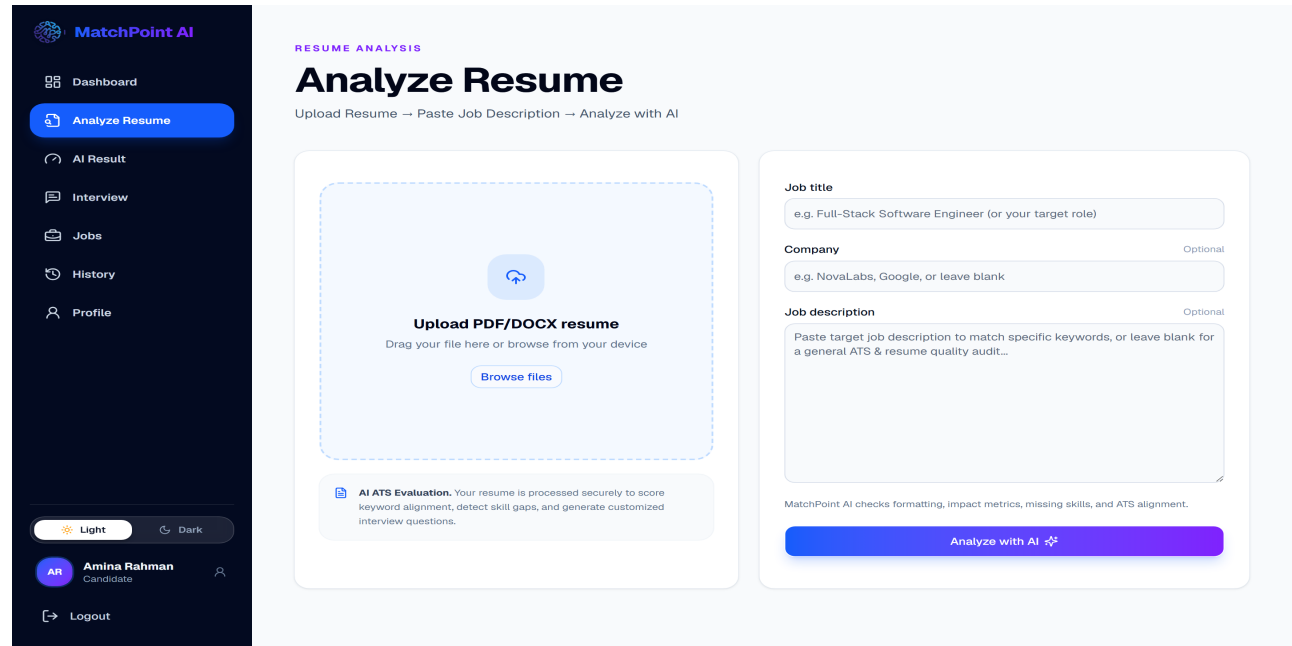
Figure 2: Candidate Dashboard Workspace



Candidate analytics, recent analysis records, score trends, and quick actions.

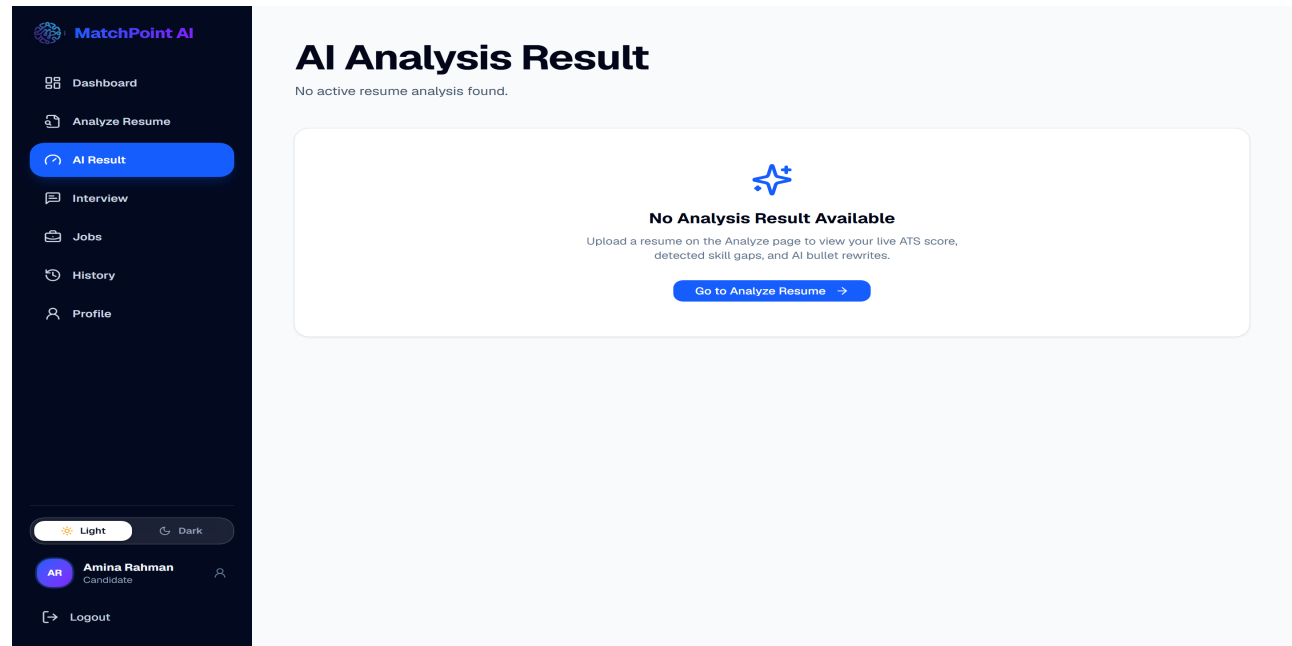
6.2 Visual Verification — Live System Screenshots (1×1 Format)

Figure 3: Resume Upload & JD Ingestion



Drag-and-drop resume upload zone with target role description form.

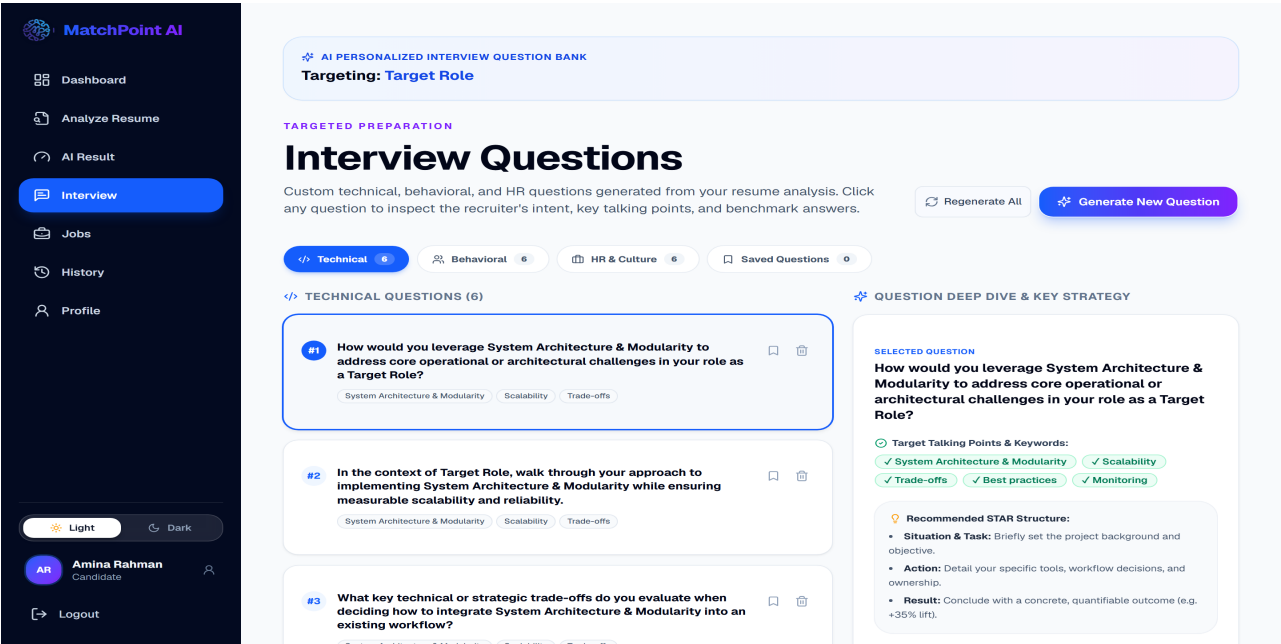
Figure 4: Radial ATS HUD & STAR Optimizer



Target HUD score gauge, keyword gap matrix, and Before/After optimizer.

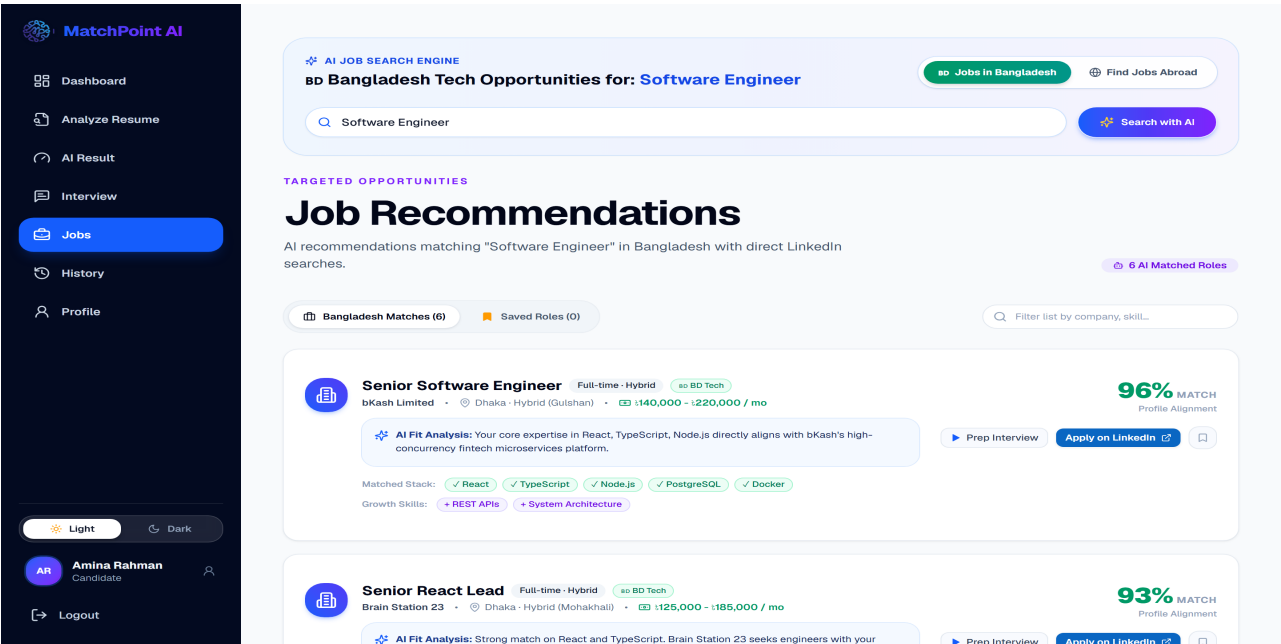
6.3 Visual Verification — Live System Screenshots (1×1 Format)

Figure 5: Tailored Interview Question Bank



Technical, Behavioral, and HR question bank with recruiter context and strategy notes.

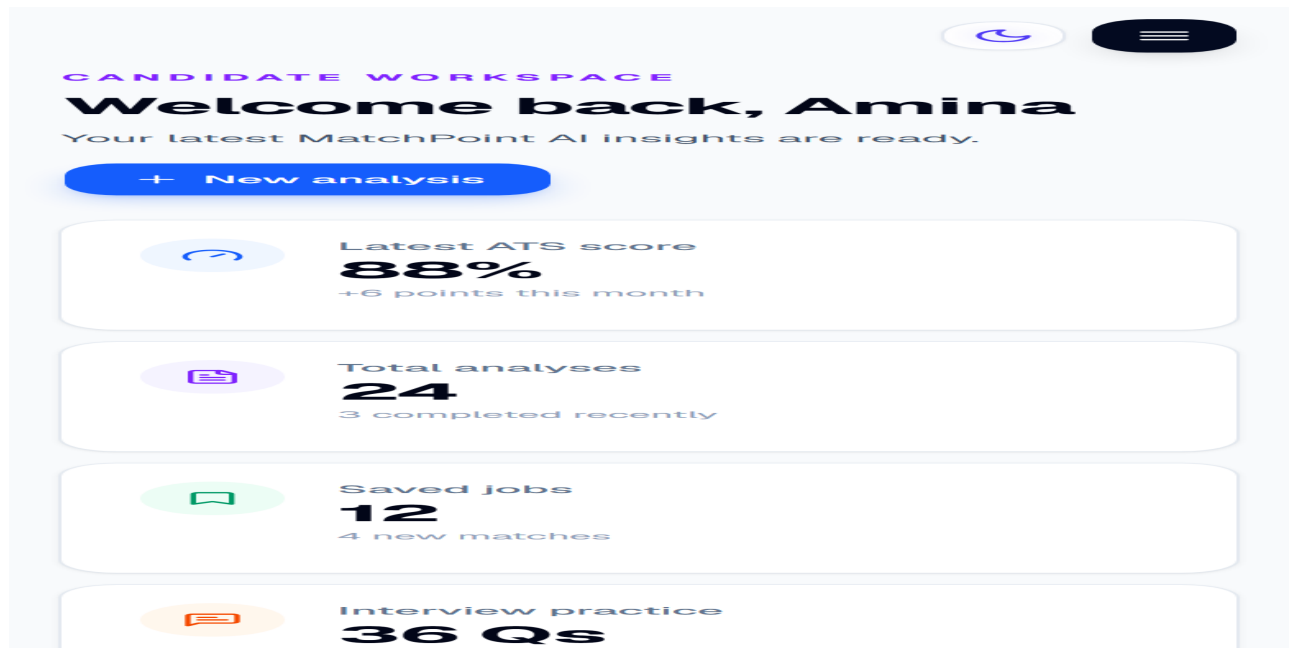
Figure 6: Dual-Market Job Matching



Dual-market job filters for Bangladesh Tech and Global Remote with 1-click LinkedIn links.

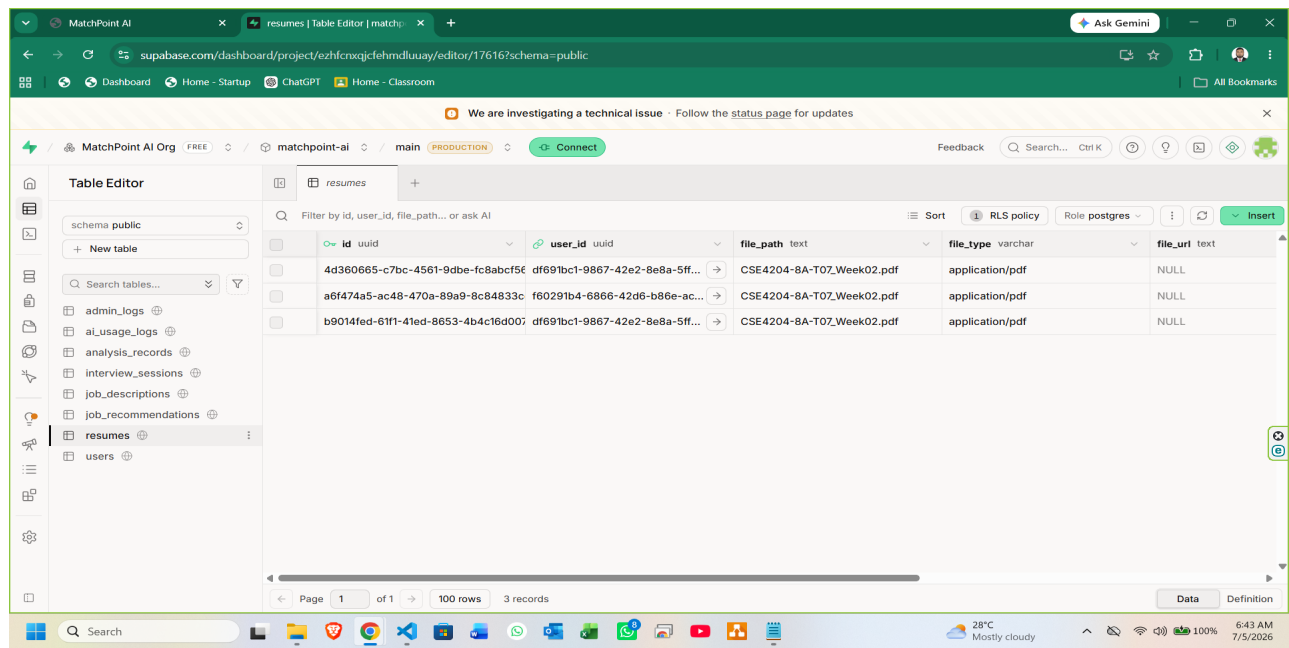
6.4 Visual Verification — Live System Screenshots (1×1 Format)

Figure 7: Mobile Responsive Interface



Responsive mobile viewport (390x844) with drawer navigation and card stacking.

Figure 8: Database & Backend API Verification



Live Supabase PostgreSQL tables, RLS policies, and Postman API runs.